

Spatial Tensor Simulation

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Abstract—TBD**Keywords**—TBD

1. SPATIAL DATA

Spatial data can be easily generated using a simple grid. For the purposes of comparing how spatial smoothers behaves we will generate a small square grid of 10×10 . The following figure shows the grid used for this simulation.

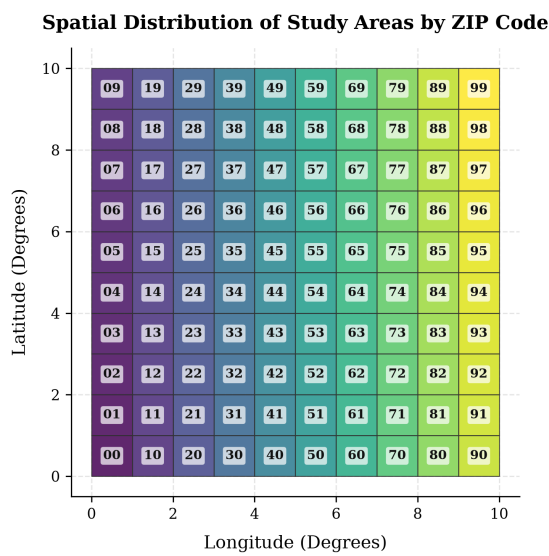


Figure 1. Simulated Zipcode Grid Map

As this not a single point in time, we look how this each spatial agent behaves through space. This this we will generate a balance panel dataset [1]

This simulated will be generate an outcome variable that it distributes from a normal distributions were the data is unfenced spatially by its noughts. This can be summarized with the following model

$$Y \sim N(\alpha + \beta_1 X_1 + \beta_2 X_2 + \rho WY) \quad (1)$$

$$X_i \sim N(2, 3), \quad \text{For } 1, 2, 3$$

Where the ρ is the auto regressive component,

REFERENCES

- [1] G. Chamberlain, "Panel data", *Handbook of econometrics*, vol. 2, pp. 1247–1318, 1984.

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